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Practical 8

Aim INTRUSION DETECTION SYSTEM

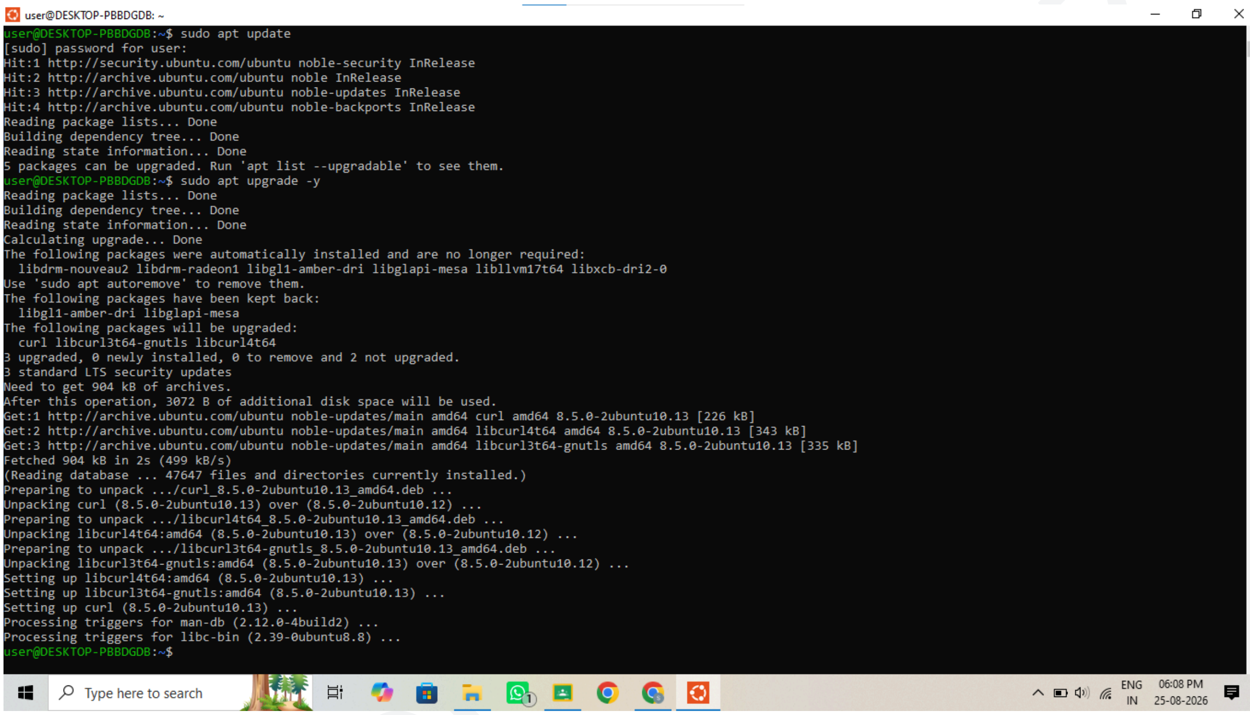
To set up and configure an open-source Intrusion Detection System (IDS) using Snort to monitor network traffic and detect potential security breaches or malicious activities.

Step 1: Update Ubuntu

Open the Terminal and run:

sudo apt update

sudo apt upgrade -y



Step 2: Install Snort

Install Snort using:

sudo apt install snort -y

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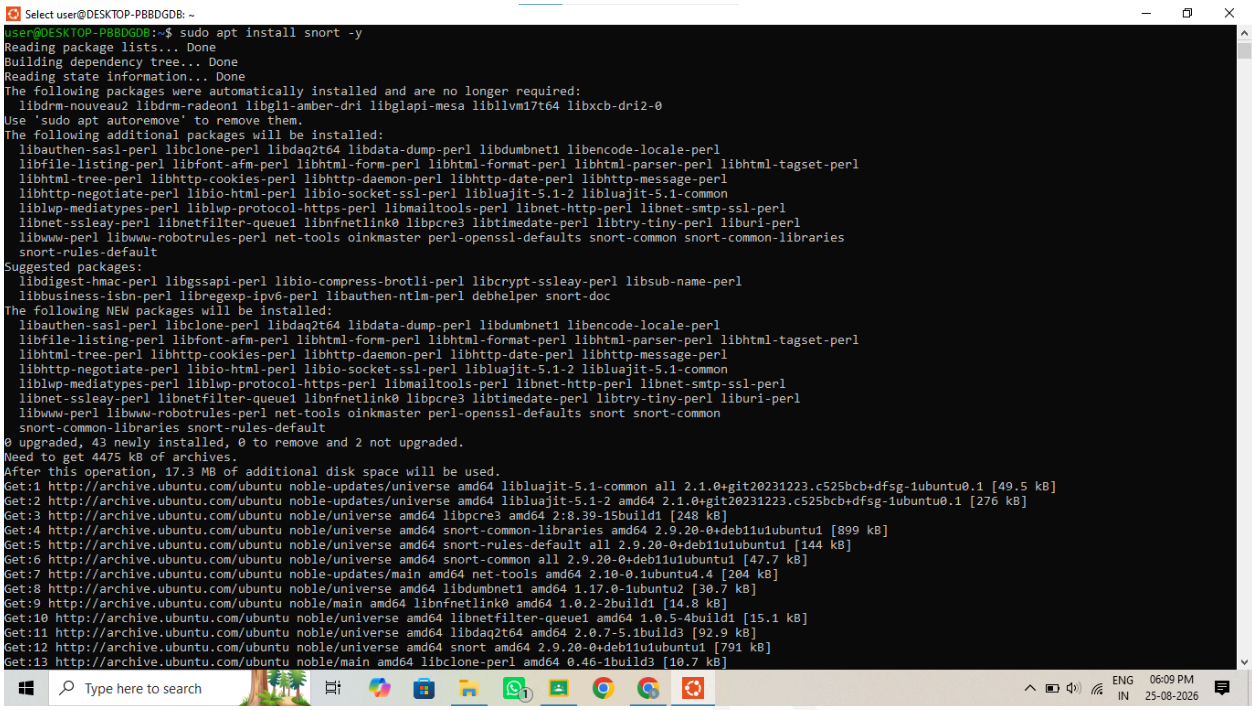
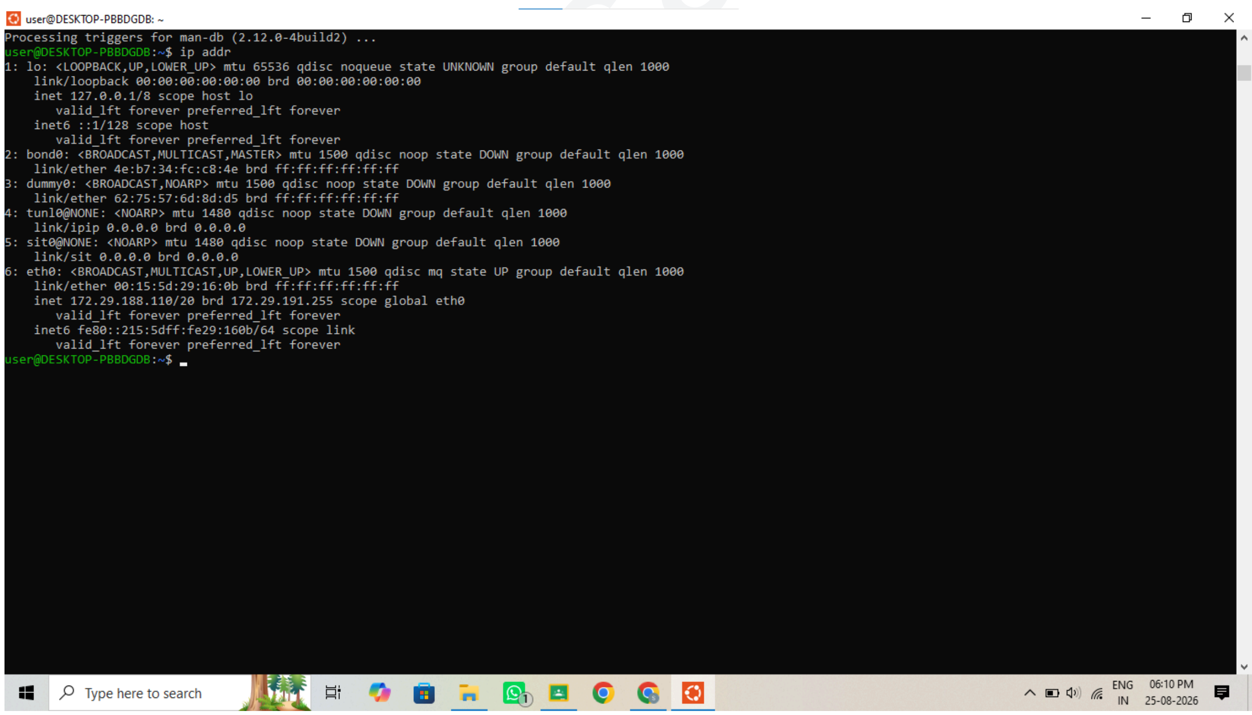


Step 3: Check the Network Interface

Use:

ip addr



Step 4: Test Snort Configuration

Run:

sudo snort -T -c /etc/snort/snort.conf

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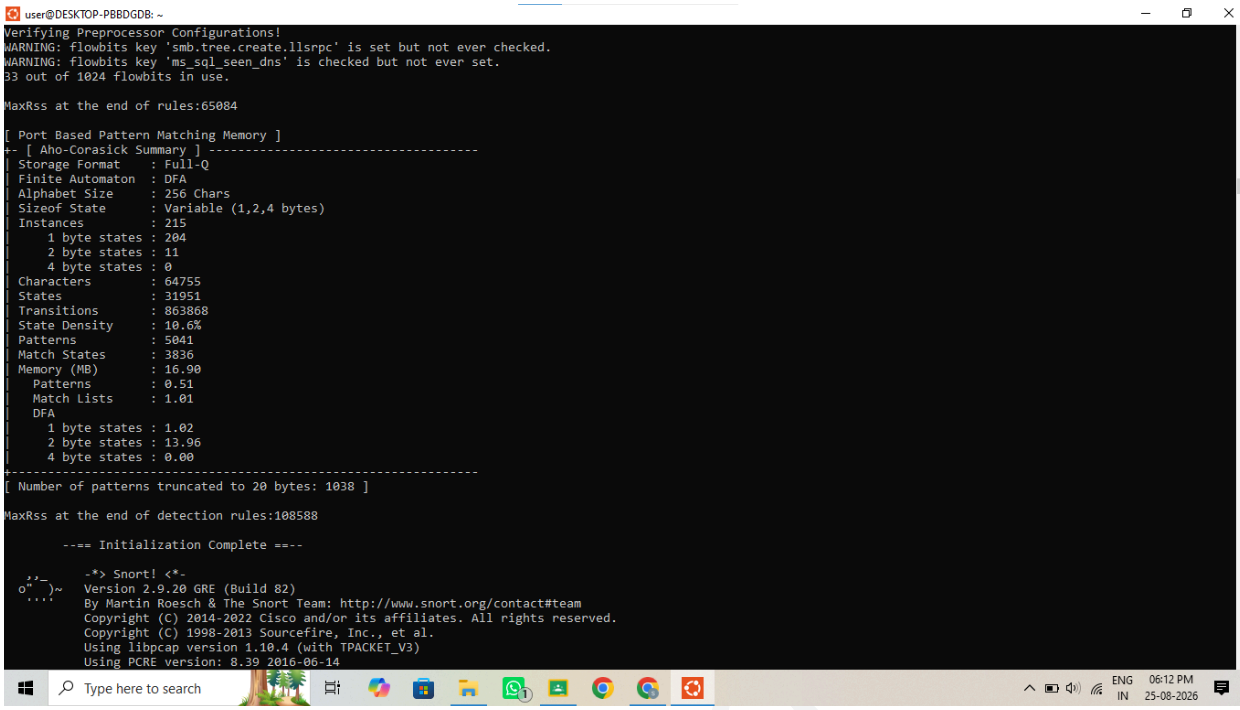

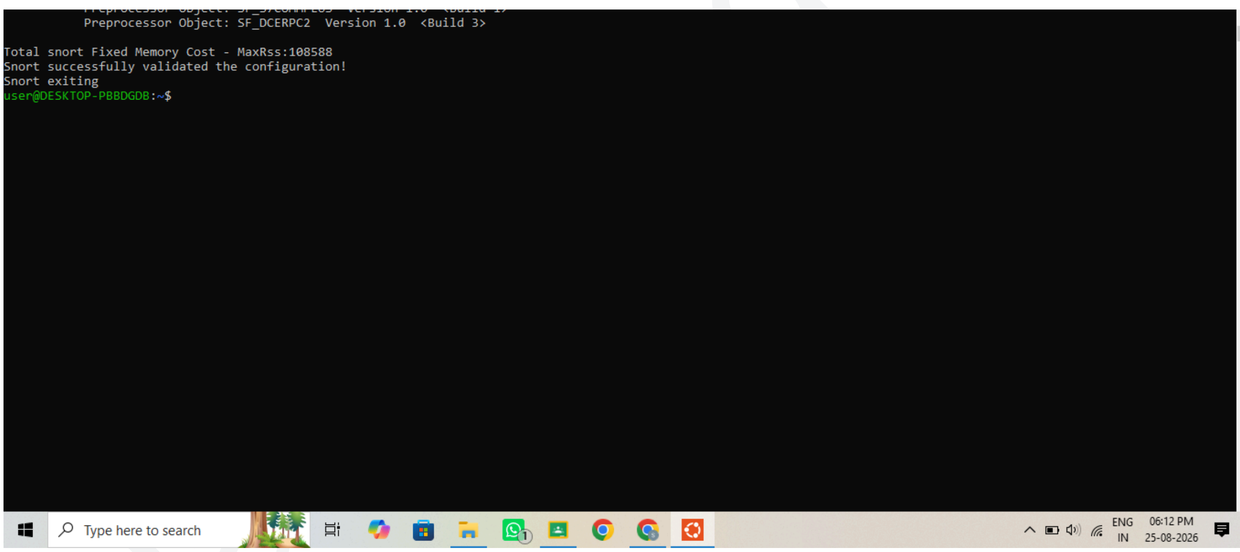

Step 5: Start Snort in Network Monitoring Mode

Run:
sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Replace enp0s3 with your actual network interface.
- -i specifies the network interface.
- -A console displays alerts on the terminal.
- -c specifies the Snort configuration file.

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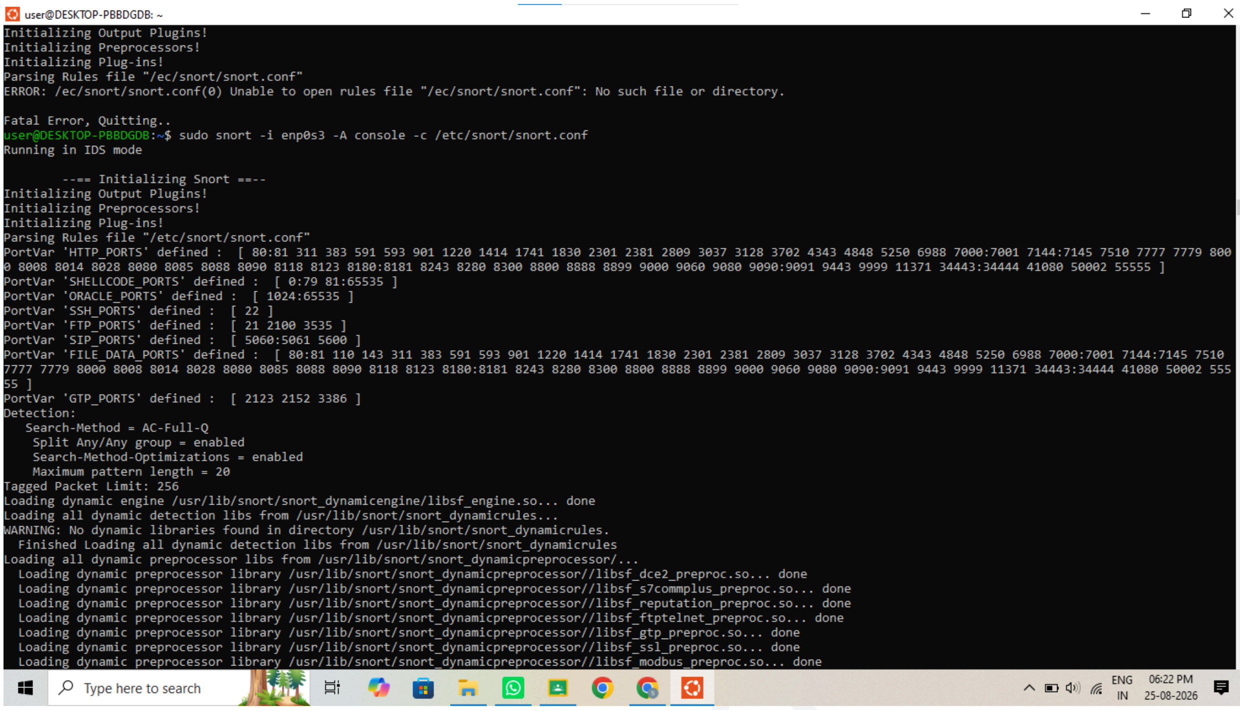


Step 6: Create a Custom IDS Rule

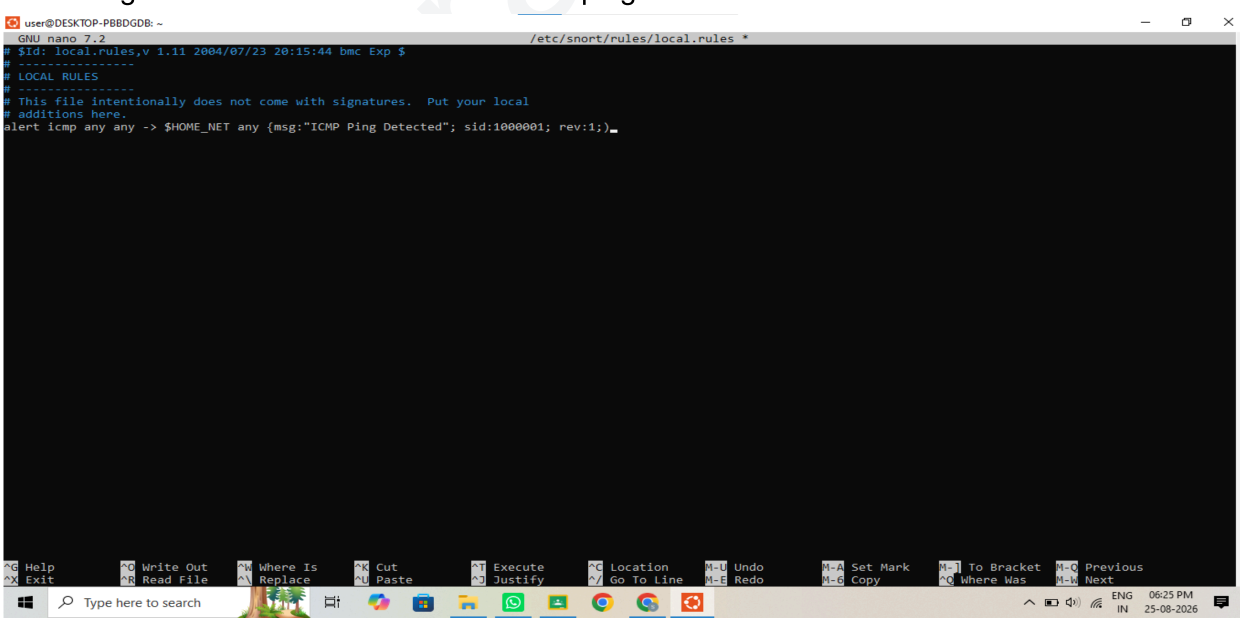
Open the local rules file:

sudo nano /etc/snort/rules/local.rules

Add:

alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

This rule generates an alert whenever ICMP ping traffic is detected.



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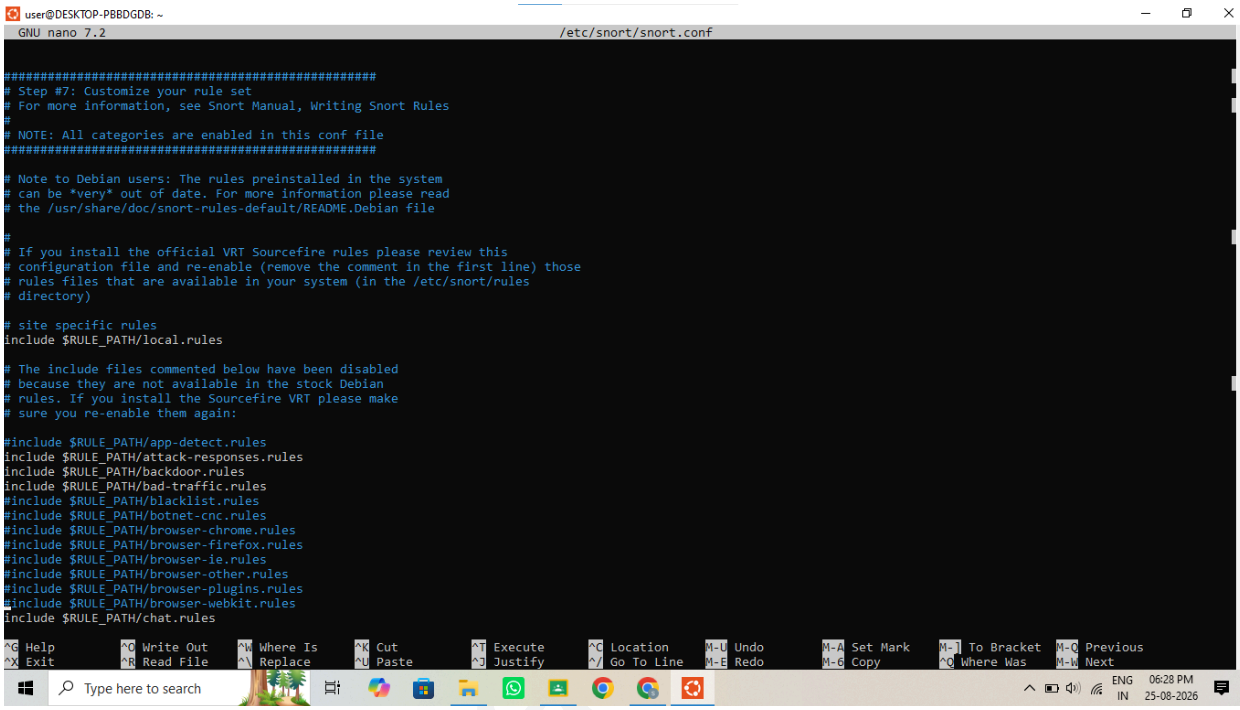
Step 7: Ensure the Local Rule File Is Included

Open:
sudo nano /etc/snort/snort.conf
Check that this line is present:
include $RULE_PATH/local.rules



Step 8: Test the Configuration Again

Run:
sudo snort -T -c /etc/snort/snort.conf
Correct any errors before continuing.

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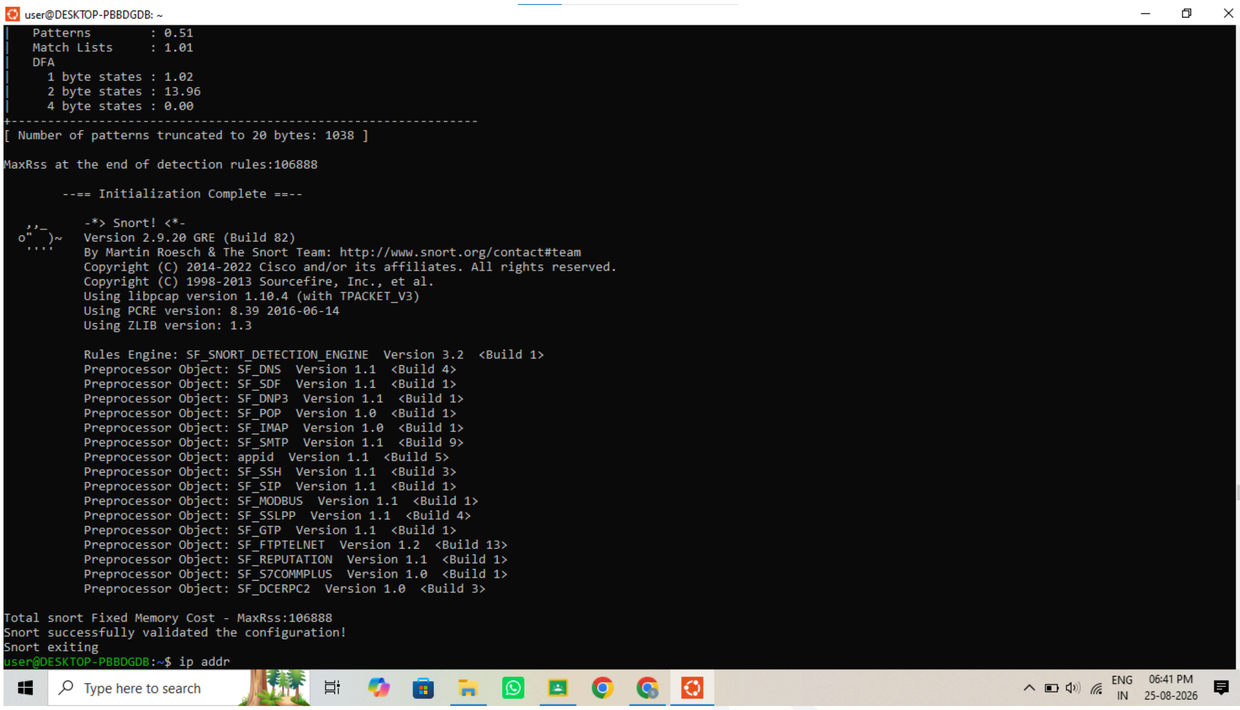
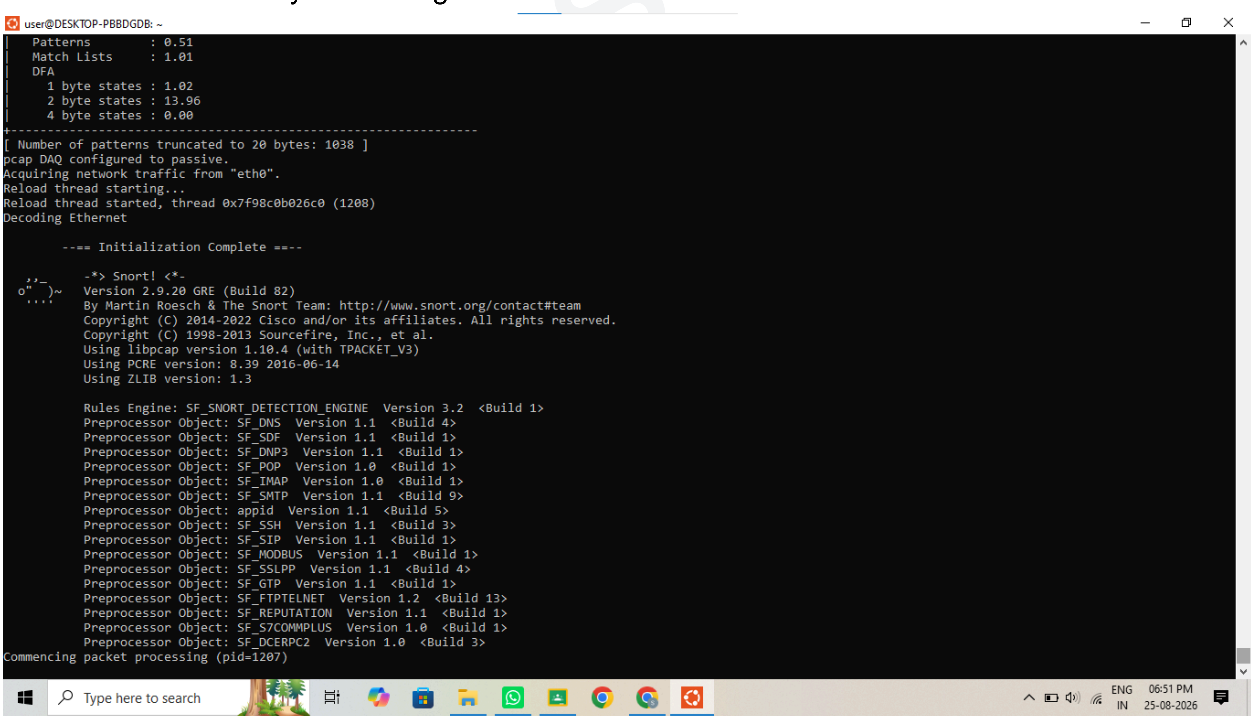


Step 9: Start IDS Monitoring

Run:

sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf

The IDS is now actively monitoring network traffic.



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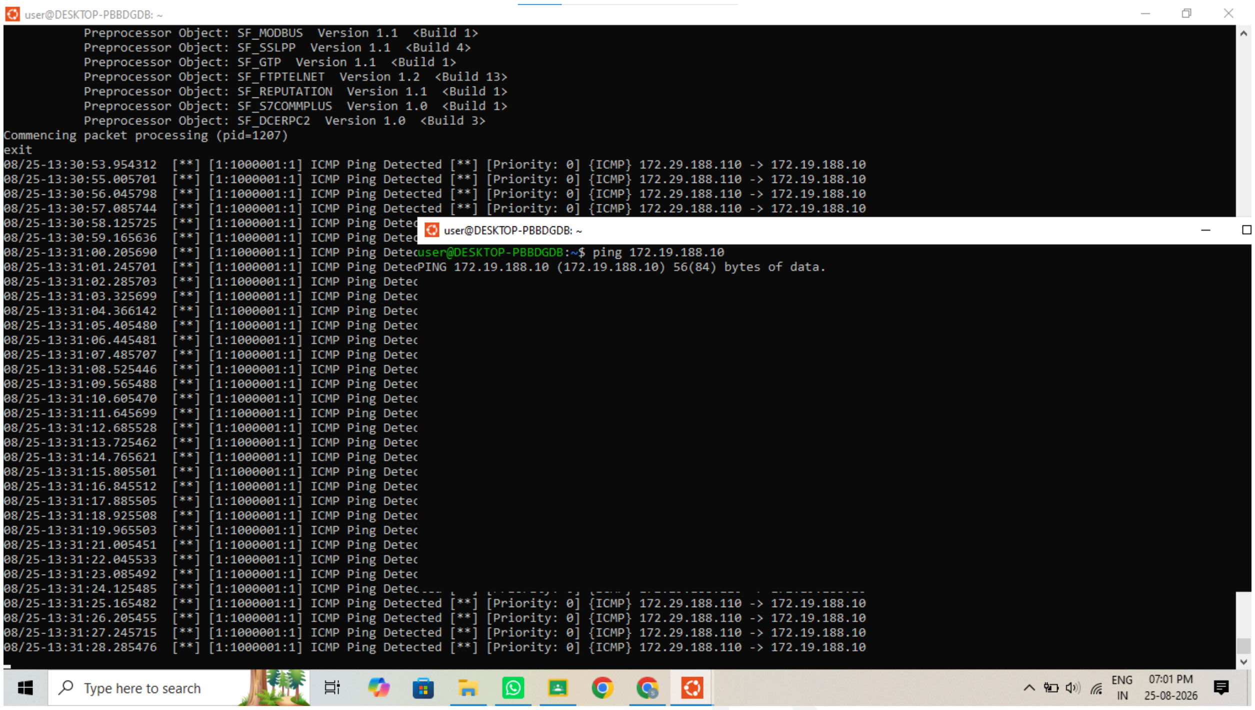


10. Step 7: Ensure the Local Rule File Is Included

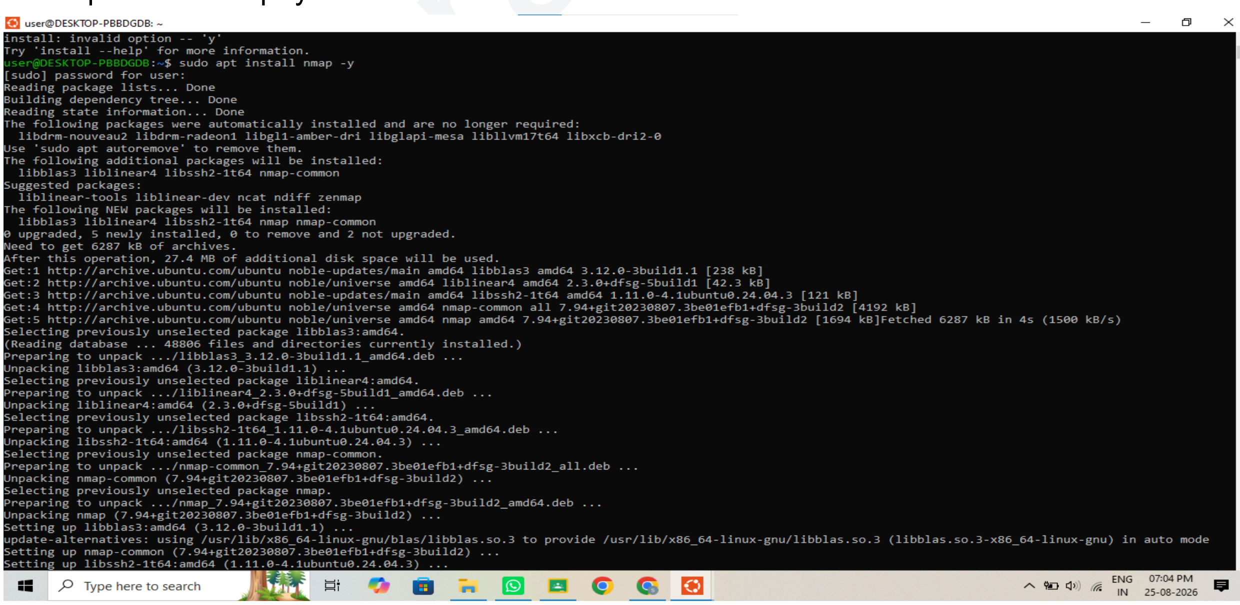
Open:

sudo nano /etc/snort/snort.conf

Step 11: Test Port Scanning Detection

Install Nmap:

sudo apt install nmap -y



Add the following rule to /etc/snort/rules/local.rules:

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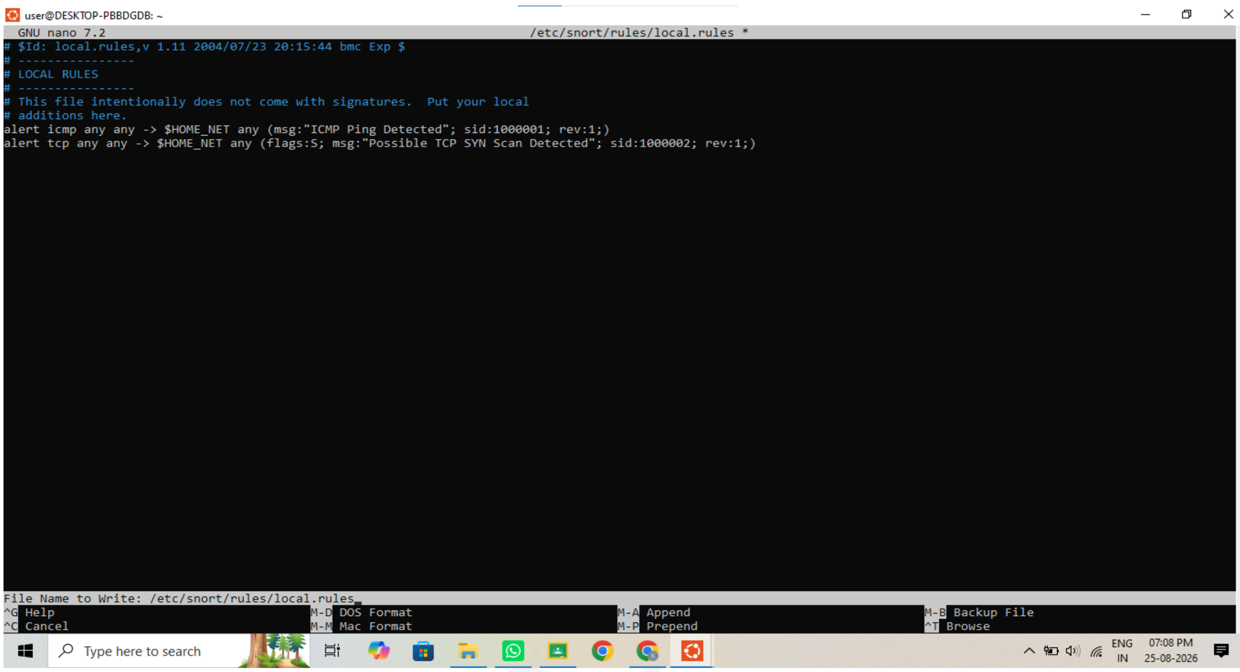
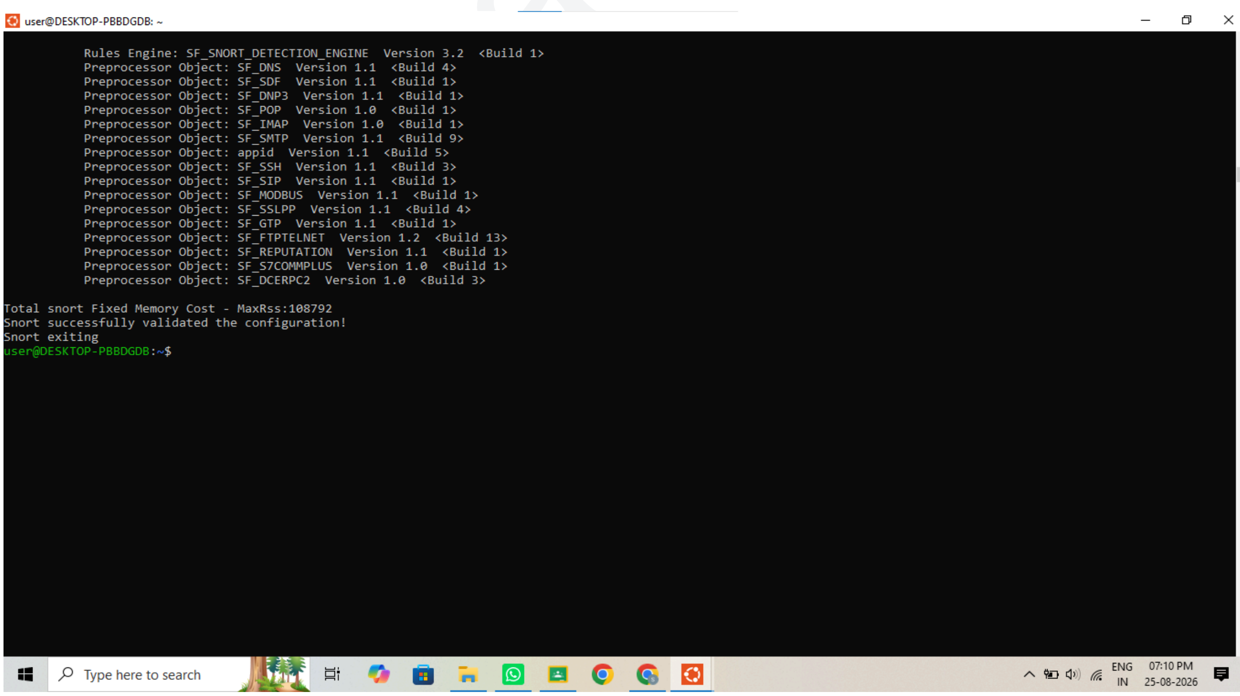
alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected";
sid:1000002; rev:1;)
Restart Snort. From an authorized lab machine:
nmap -sS <TARGET-IPADDRESS>





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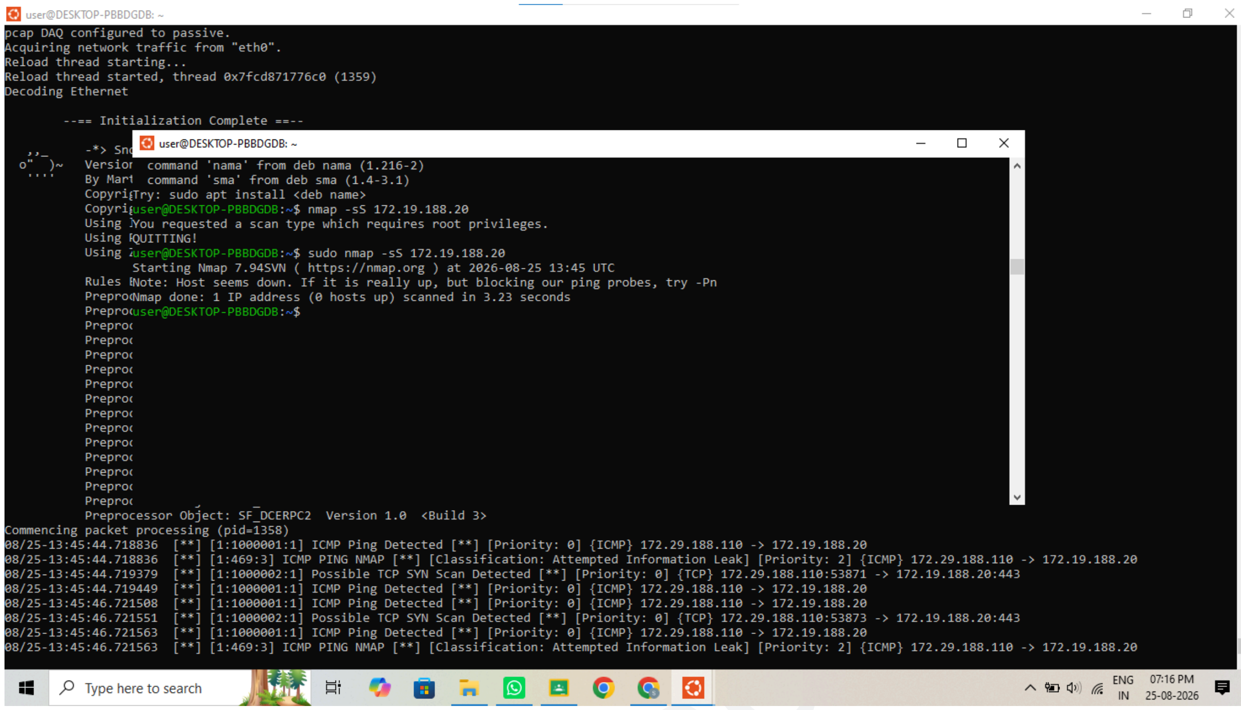
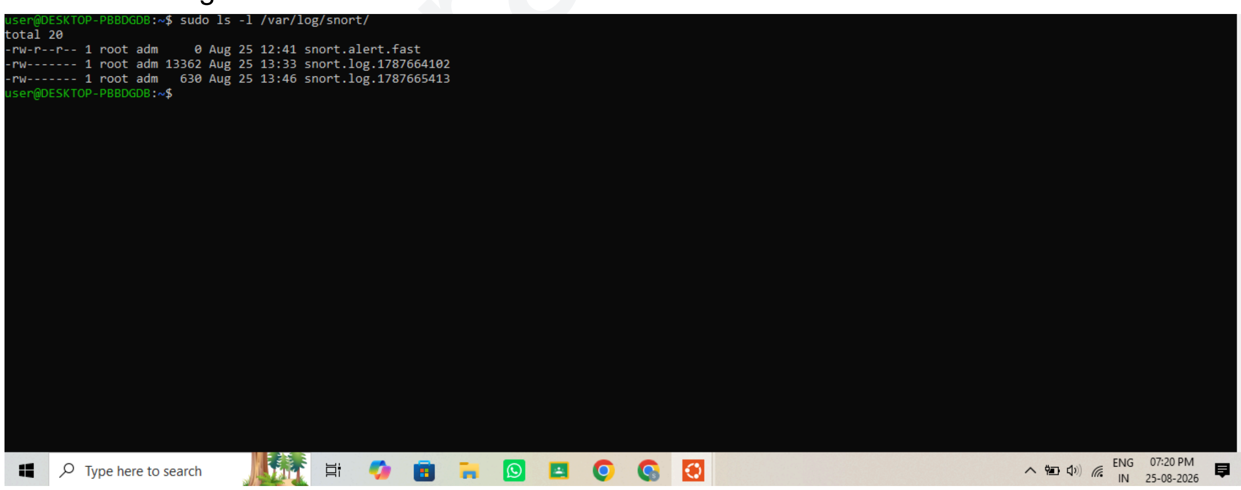


Step 12: View IDS Alerts

Alerts may be displayed directly when using -A console. Logs can also be checked in:
/var/log/snort/
Commands:
sudo ls /var/log/snort/
sudo cat /var/log/snort/alert



Step 13: Monitor Traffic Using Wireshark

Install Wireshark:
sudo apt install wireshark -y

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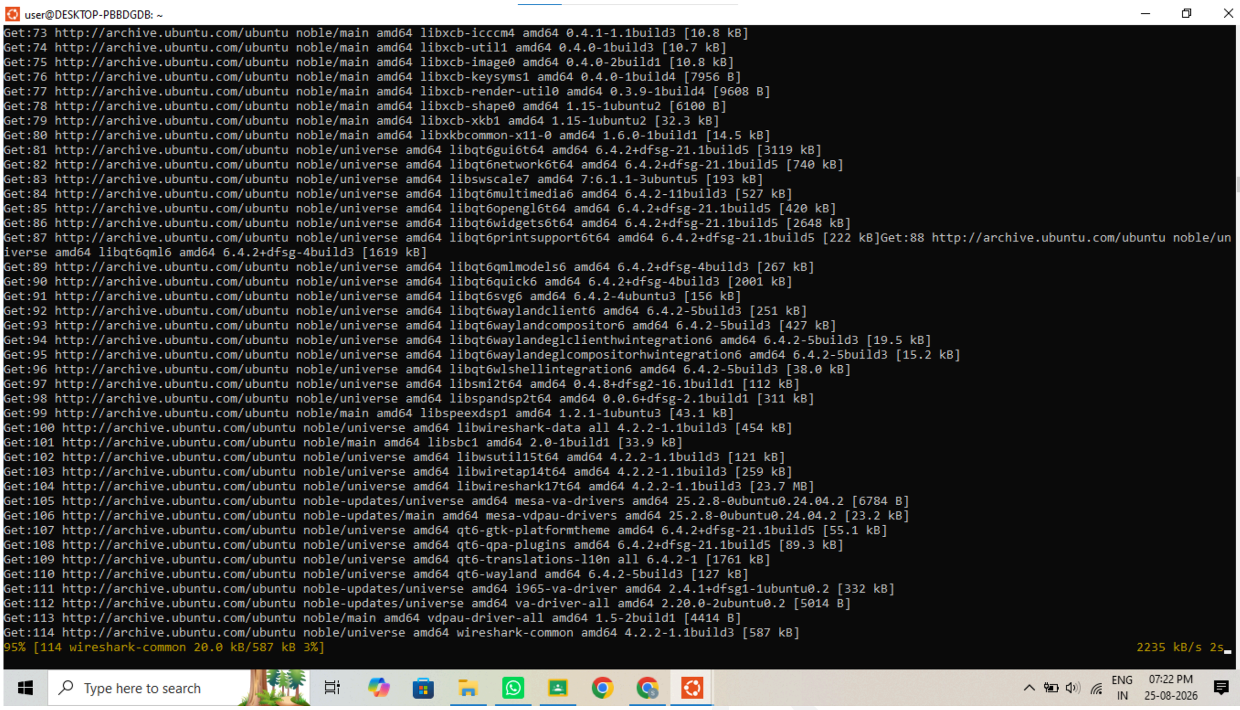

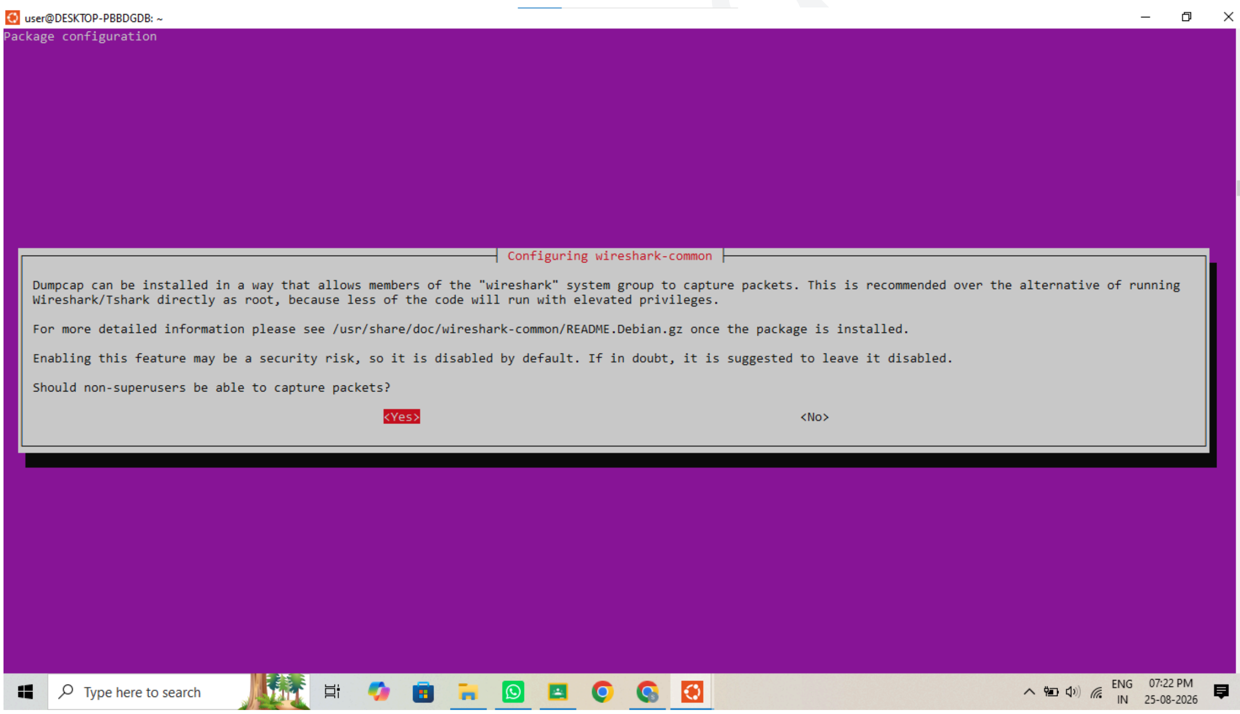

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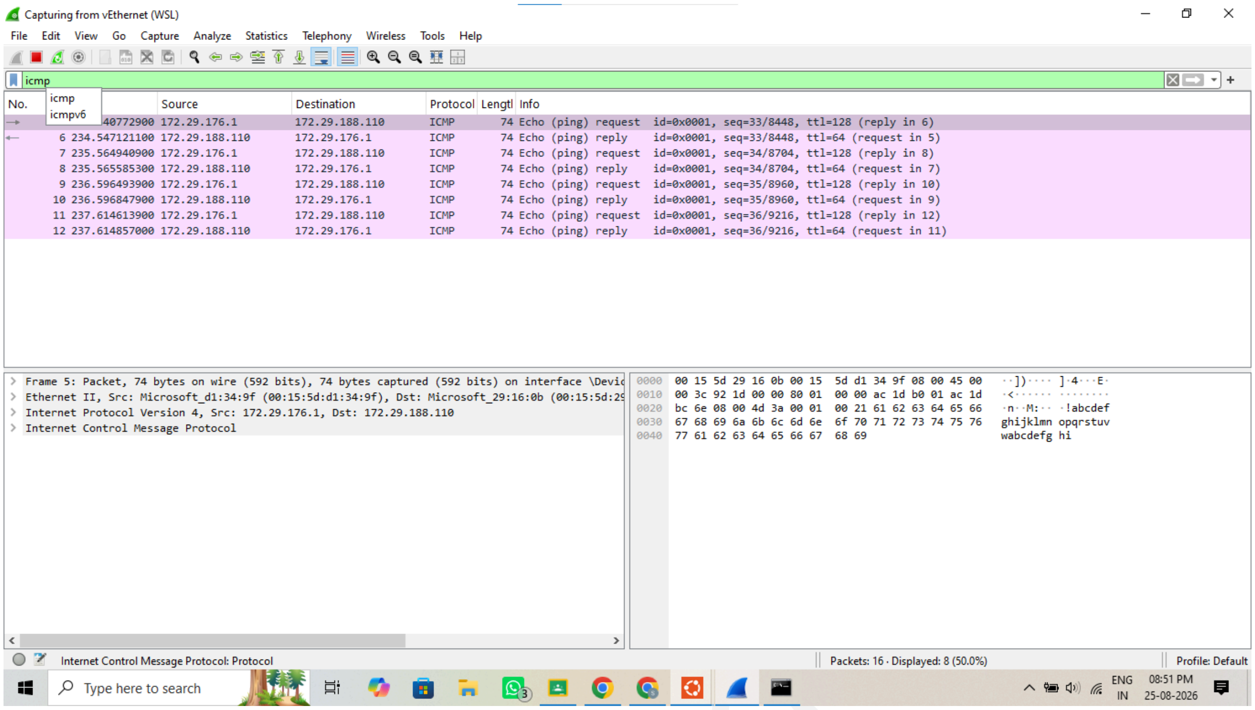

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